

Nordhaus–Gaddum Bounds for Independent Domination

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Abstract

We establish sharp bounds on the sum and product of the independent domination numbers of a graph and its complement.

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The independent domination number $i(G)$ of a graph G is the minimum cardinality of a set that is both independent and dominating. Equivalently, it is the minimum cardinality of a maximal independent set.

Cockayne and Mynhardt [2] explored the sum and product of the independent domination numbers of a graph and its complement. In particular, they found a graph for order p a multiple of 4 where $i(G)i(\overline{G}) = (p + 4)^2/16$. Later with Fricke [3], they showed that this is best possible in that $i(G)i(\overline{G}) \leq p^2/16 + o(p^2)$. We show that the original value is exactly the maximum possible.

They (and others) observed that the maximum value of $i(G) + i(\overline{G})$ is $p + 1$, attained for example for the complete graph. But the question of the maximum when neither G nor $\overline{G}$ has an isolated vertex remained open. This we answer too. See also the discussion in Section 9.1 of [7] where these questions are listed as open problems. For a survey of some related problems see [6].

The key is the result of Entringer et al. [4]

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Theorem 1 [4] *For $m, n \geq 2$, the minimum order of a graph such that every vertex is in an m -clique and an independent set of order n is $\lceil (\sqrt{m-1} + \sqrt{n-1})^2 \rceil$.*

We will need the following extremal graphs. For $d \leq s$, let graph $G(r, s, d)$ be a graph consisting of the disjoint union of a clique R on r vertices and an independent set S of s vertices with edges added such that every vertex in R is adjacent to d vertices in S and every vertex in S is adjacent to either $\lfloor rd/s \rfloor$ or $\lceil rd/s \rceil$ vertices in R . Then:

$$i(G(r, s, d)) = s - d + 1 \quad \text{and} \quad i(\overline{G}(r, s, d)) = \lfloor rd/s \rfloor + 1.$$

Now define:

$$b(p) = \left\lfloor \frac{p+4}{4} \right\rfloor \left\lfloor \frac{p+6}{4} \right\rfloor.$$

That is, $b(p)$ is $(p+4)^2/16$ if $p \equiv 0 \pmod{4}$, $(p+3)^2/16$ if $p \equiv 1 \pmod{4}$, $(p+2)(p+6)/16$ if $p \equiv 2 \pmod{4}$, and $(p+1)(p+5)/16$ if $p \equiv 3 \pmod{4}$.

Theorem 2 *For all graphs G of order p ,*

$$i(G)i(\overline{G}) \leq \begin{cases} p & \text{if } p \leq 7 \\ b(p) + 1 & \text{if } p = x^2 \text{ for } x \text{ odd, or } p = x^2 - 1 \text{ for } x \text{ even} \\ b(p) & \text{otherwise,} \end{cases}$$

and this is best possible for all p .

PROOF. Let $i(G) = m$ and $i(\overline{G}) = n$. If $m = 1$ or $n = 1$, then the maximum product is clearly p .

So assume $m, n \geq 2$. (In particular, $p \geq 4$.) Then every vertex is in an independent set of size m and a clique of size n . Thus by Theorem 1,

$$p \geq (\sqrt{m-1} + \sqrt{n-1})^2.$$

Consider then the problem of maximising mn subject to the above constraint and $m, n \geq 2$. If we set $\mu = \sqrt{m-1}$, $\nu = \sqrt{n-1}$ and $\rho = \sqrt{p}$, we get:

$$\text{maximise: } E = (\mu^2 + 1)(\nu^2 + 1) \text{ such that } \mu + \nu \leq \rho \text{ and } \mu, \nu \geq 1.$$

Clearly the maximum has $\nu = \rho - \mu$. A little calculus shows that the global maximum for $E = (\mu^2 + 1)((\rho - \mu)^2 + 1)$ is $E^* = (p/4 + 1)^2$ attained for $\mu = \rho/2$.

That is, the extremum is attained for $m = n = p/4 + 1$, at least for p a multiple of 4. If p is even, but not a multiple of 4, then the value $\lfloor (p/4 + 1)^2 \rfloor$ is attainable (as the extremal graphs given below show).

When p is odd, the story is a bit more complicated due to integrality. Consider first $p = 4r + 1$. If we set $a = m - 1$ and $b = n - 1$, and rearrange the condition, we need to:

maximise $E(a, b) = (a + 1)(b + 1)$ subject to $C(a, b) = (p - a - b)^2 - 4ab \geq 0$ with $1 \leq a, b \leq p - 1$.

Note that the function $E(a, b)$ is increasing in both a and b while the function $C(a, b)$ is decreasing in both a and b (in this range).

Furthermore, $C(r + 1, r + 1) = -12r - 3$. Therefore, for the condition $C \geq 0$ to be satisfied, at least one of a, b , say a , is at most r . Let $a = r - u$ for integer $u \geq 0$. Since $C(r - u, r + u) = 1 + 4r + 4u^2 > 0$, we may assume that $b \geq r + u$. Let $b = r + u + e$ for integer $e \geq 0$.

If $e = 0$ then $E = (r + 1)^2 - u^2$ which attains its largest value for $u = 0$. The values $u = e = 0$ (meaning $a = b = r$) satisfy the condition (since $C(r, r) = 1 + 4r$); thus the maximum is at least $(r + 1)^2$. We shall now examine if $e \geq 1$ can give a larger value.

Consider a new expression F given by

$$F = 4(E - (r + 1)^2) + C = (e + 1)^2 - 4r(e - 1).$$

If $E > (r + 1)^2$ and $C \geq 0$, then $F > 0$. So we can restrict attention to those values of e and r where $F > 0$ and hence by integrality $F \geq 1$.

Note that $e = b - r - u \leq 3r$. By a calculation, $F \geq 1$ and $e \leq 3r$ implies that $e \leq 10$ and $r \leq 10/3$ (and thus $p \leq 13$). By a further calculation, if one insists that $e \geq 2$ and both e and r integral, then it can be checked that there are only 8 possible pairs, and for all these the value of C is negative. Hence we can restrict attention to where $e = 1$.

If we set $\alpha = r - u - u^2$, then $C = -4\alpha$ and $E - (r + 1)^2 = 1 + \alpha$. Since α is an integer, the only way it can happen that $E > (r + 1)^2$ and $C \geq 0$ is for $\alpha = 0$. The equation $r - u - u^2 = 0$ has solution $(-1 \pm \sqrt{4r + 1})/2$, and so has an integer solution if and only if $4r + 1$ is a perfect square. In this case $E = (r + 1)^2 + 1$.

Thus we have shown that the maximum value of E is $(r+1)^2$, unless $4r+1$ is a perfect square in which case the maximum is $(r+1)^2 + 1$.

The method for $p = 4r-1$ is the same and we omit the details. (As $C(r, r) < 0$ while $C(r-u-1, r+u) > 0$, we set $a = r-u-1$ and $b = r+u+e$ for integers $e, u \geq 0$. The maximum for $e = 0$ is $r(r+1)$, and again we show that for $e \geq 2$ this cannot be exceeded, while for $e = 1$ this is exceeded iff r is a perfect square.)

The extremal graphs are as follows (we omit the details of the calculations).

- K_p for $p \leq 7$;
- for $p = x^2$ for x odd, it is $G(x(x+1)/2, x(x-1)/2, (x^2-1)/4)$; (here $i(G) = (x-1)^2/4 + 1$ and $i(\overline{G}) = (x+1)^2/4 + 1$);
- for $p = (2y)^2 - 1$ for y integer, it is $G((y+1)(2y-1), y(2y-1), y^2)$; (here $i(G) = y^2 - y + 1$ and $i(\overline{G}) = y^2 + y + 1$);
- and $G(\lfloor p/2 \rfloor, \lceil p/2 \rceil, \lceil p/4 \rceil)$ otherwise, where $\lceil p/2 \rceil$ denotes scientific rounding. (That is, $\lfloor x \rfloor$ is rounding to the nearest integer but if x is halfway between two integers then it is rounded to the even one.) Some calculations show that $i(G) = \lfloor (p+4)/4 \rfloor$ and $i(\overline{G}) = \lfloor (p+6)/4 \rfloor$.

□

Theorem 3 *For all graphs G of order p without isolated or dominating vertices,*

$$i(G) + i(\overline{G}) \leq p + 4 - 2\sqrt{p},$$

and this is best possible (if one rounds down) for all p (necessarily $p \geq 4$).

PROOF. Like above, the problem of maximising $m+n$ subject to the above constraint and $m, n \geq 2$ is equivalent to

$$\text{maximise: } F = (\mu^2 + 1) + (\nu^2 + 1) \text{ such that } \mu + \nu \leq \rho \text{ and } \mu, \nu \geq 1.$$

Again the maximum has $\nu = \rho - \mu$. This means that F as a function of μ is a parabola and therefore has an endpoint maximum. Clearly the two endpoints (corresponding to $\mu = 1$ and $\nu = 1$) yield the same value, and $F = p + 4 - 2\sqrt{p}$, hence the bound.

The extremal graphs are $H_p = G(\lfloor \sqrt{p} \rfloor, p - \lfloor \sqrt{p} \rfloor, \lceil \sqrt{p} - 1 \rceil)$, where tedious calculations show that $i(H_p) = \lfloor p + 2 - 2\sqrt{p} \rfloor$ and $i(\overline{H_p}) = 2$. □

As an immediate consequence of Theorem 3, we have an upper bound for the

independent domination number of a graph due to Gimbel and Vestergaard [5] (see also Theorem 10.25 in [1]).

Corollary 4 [5] *If G is a graph of order $p \geq 2$ with no isolated vertex, then $i(G) \leq p + 2 - 2\sqrt{p}$.*

PROOF. If G has a dominating vertex, then $i(G) = 1$ and the desired result follows. If G has no dominating vertex, then $p \geq 4$ and $i(\overline{G}) \geq 2$. Thus, by Theorem 3, $i(G) \leq p + 4 - 2\sqrt{p} - i(\overline{G}) \leq p + 2 - 2\sqrt{p}$. $\square$

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